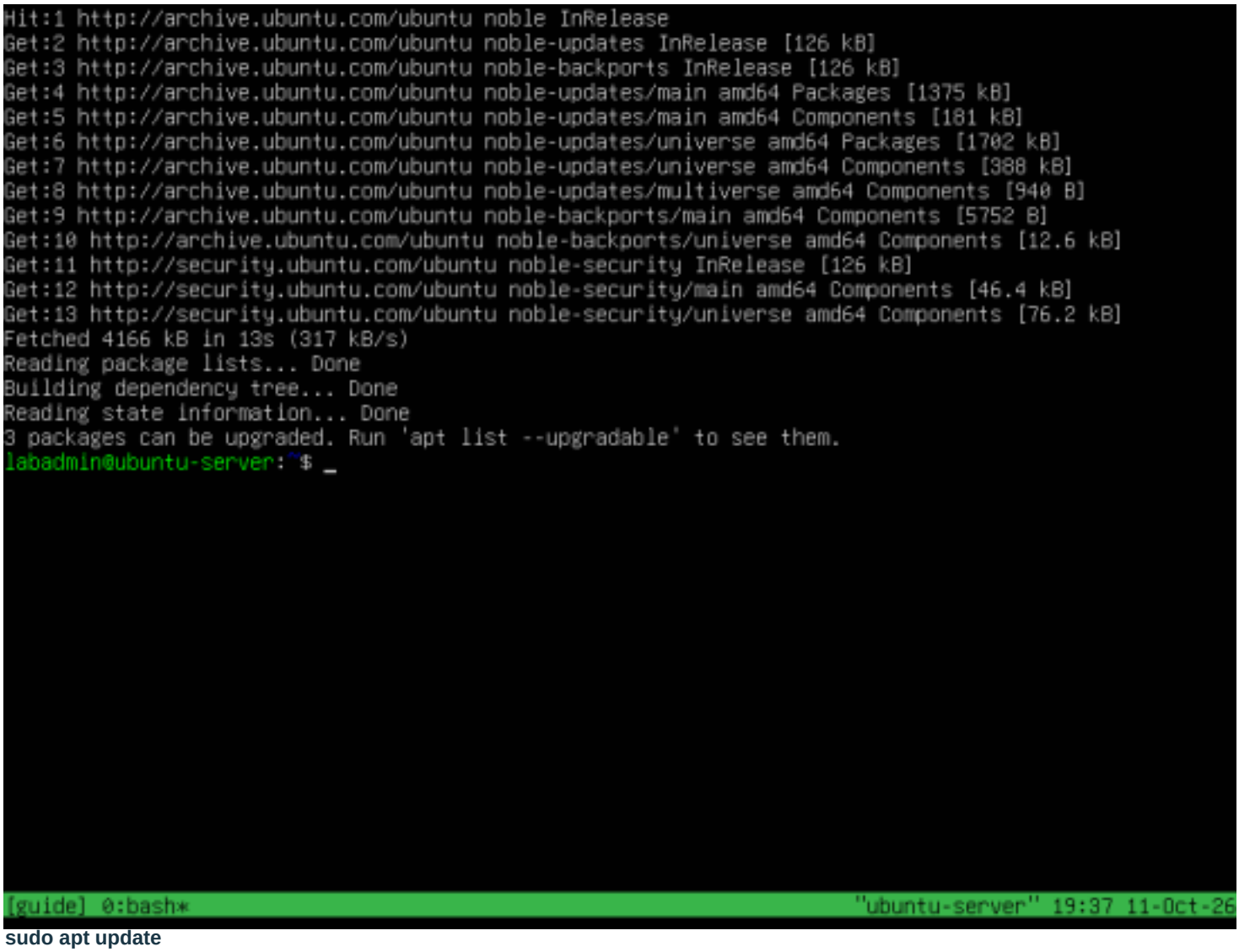
Laboratory Guide for Week 7

(IT 123 – System Administration and Maintenance)

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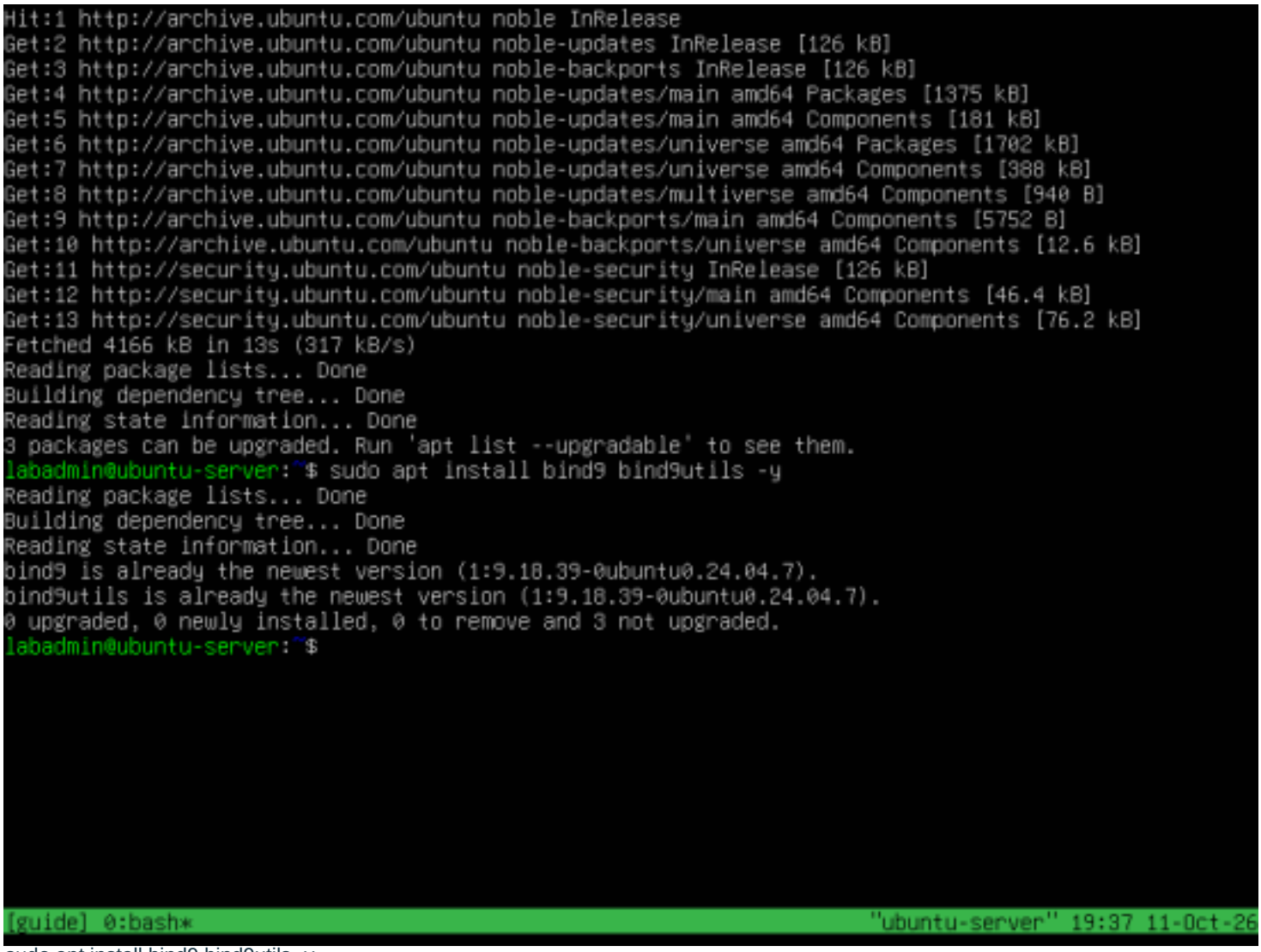
Part 1.1 - Update Package Information



sudo apt update

Ubuntu's package index was refreshed successfully. The screenshot shows the repository information being downloaded and the command finishing. Three available upgrades were reported, but this step did not ask for them to be installed.

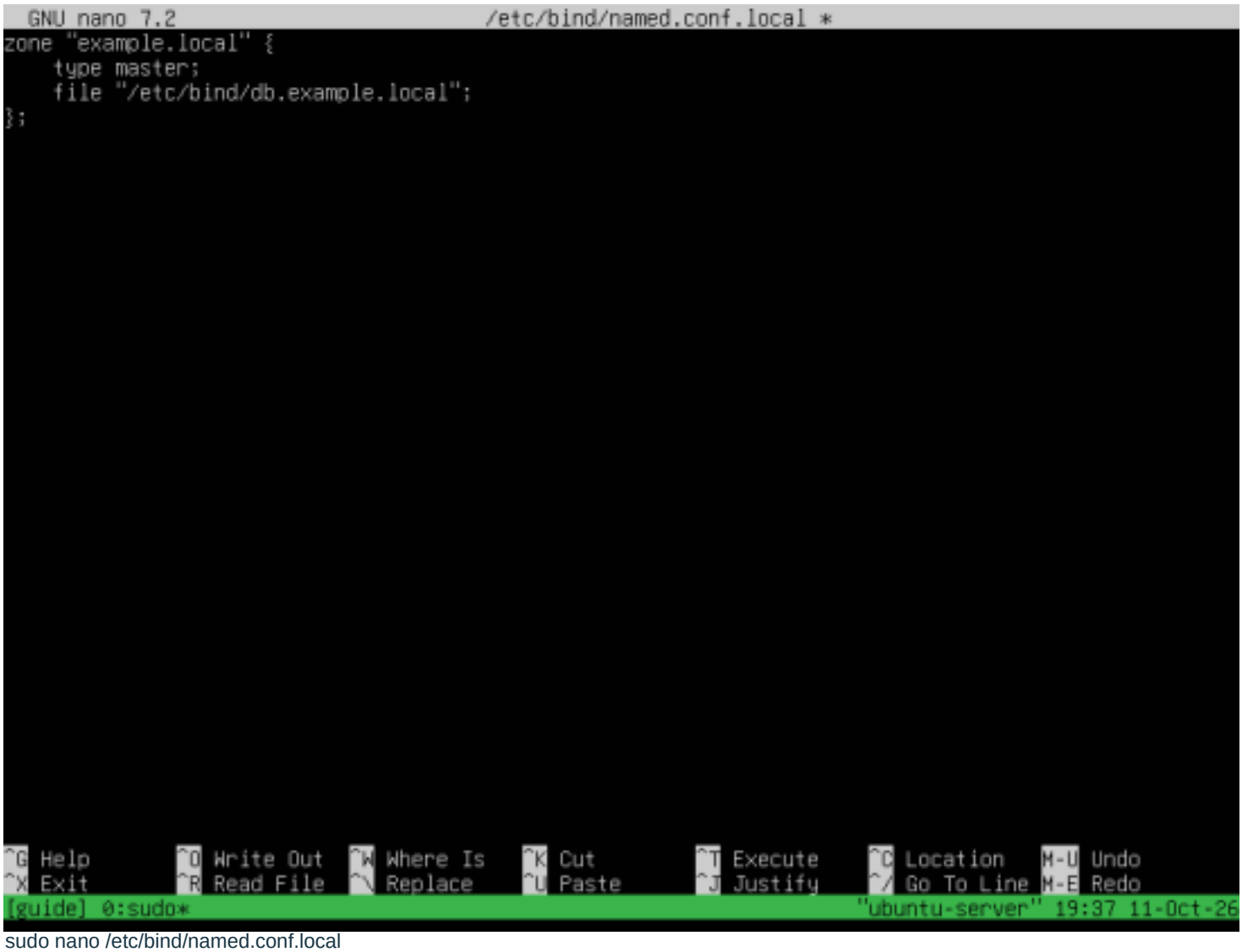
Part 1.1 - Install BIND9



sudo apt install bind9 bind9utils -y

The installation command was entered as shown. Since BIND9 and bind9utils were already installed at their latest available versions, apt did not need to install them again.

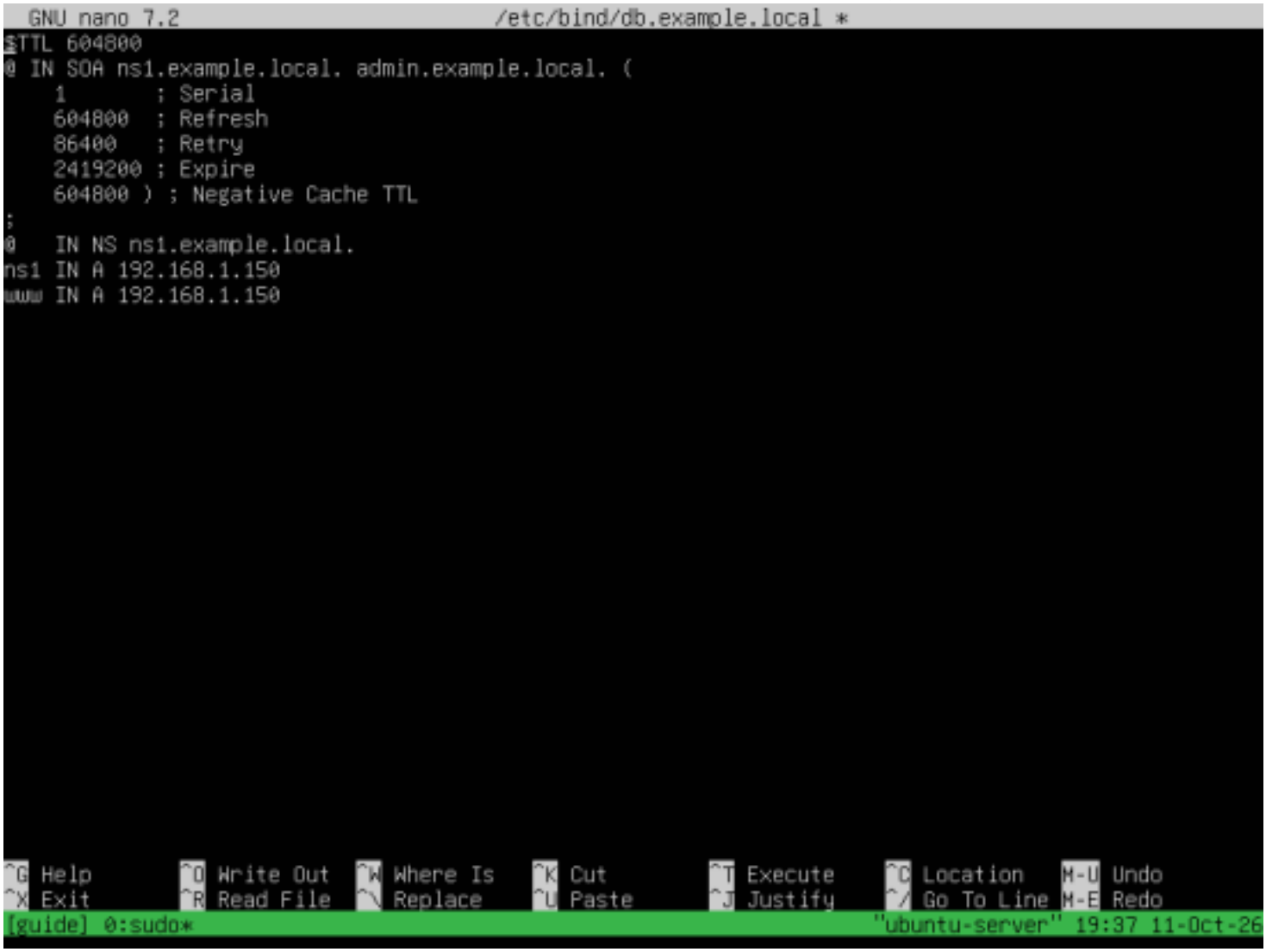
Part 1.2 - Configure the zone



sudo nano /etc/bind/named.conf.local

The screenshot shows the example.local master-zone entry in nano, with the database file set to /etc/bind/db.example.local as specified in the guide. The changes were saved with Ctrl+O, confirmed with Enter, and nano was closed using Ctrl+X.

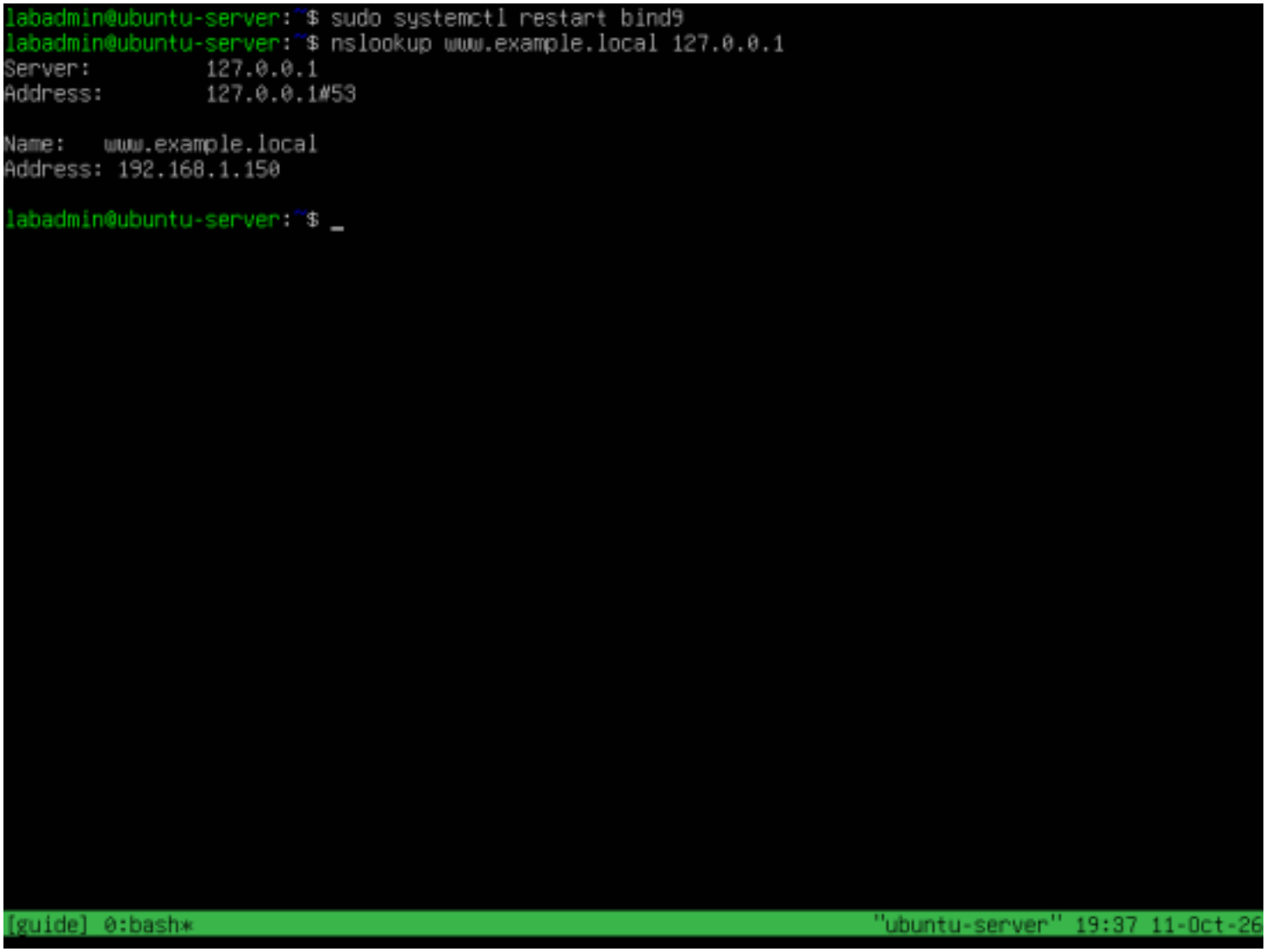
Part 1.3 - Create the zone database



sudo cp /etc/bind/db.local /etc/bind/db.example.local
sudo nano /etc/bind/db.example.local

The db.local template was copied and edited in nano. The SOA record, serial number 1, timing settings, and NS record follow the guide. The ns1 and www records both use 192.168.1.150, and the original $TTL line remains. Although the screenshot was taken before saving, the file was saved afterward.

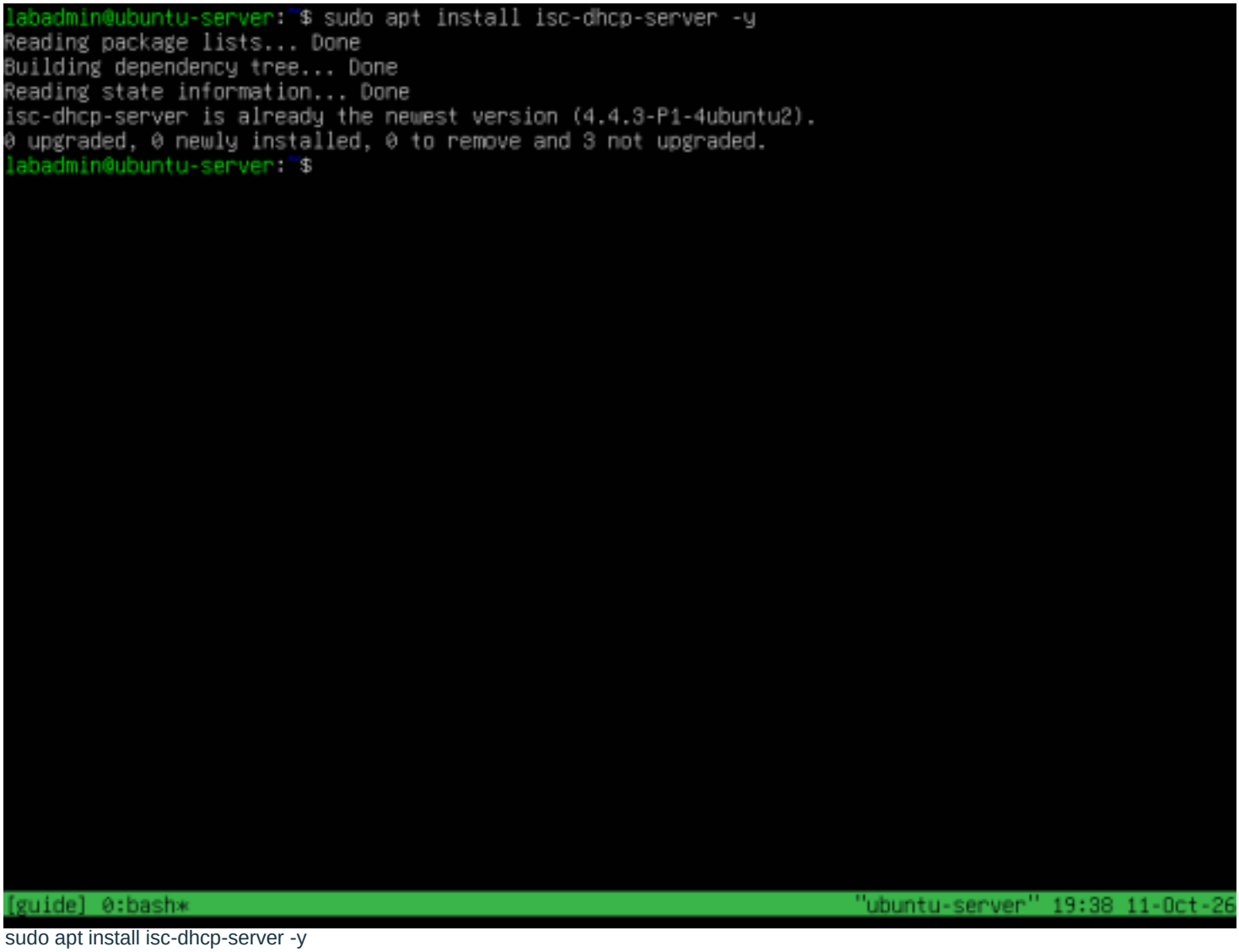
Part 1.4-5 - Restart and test DNS



sudo systemctl restart bind9
nslookup www.example.local 127.0.0.1

BIND9 restarted without displaying an error. The lookup sent to 127.0.0.1 returned 192.168.1.150 for www.example.local, matching the DNS check required by the guide.

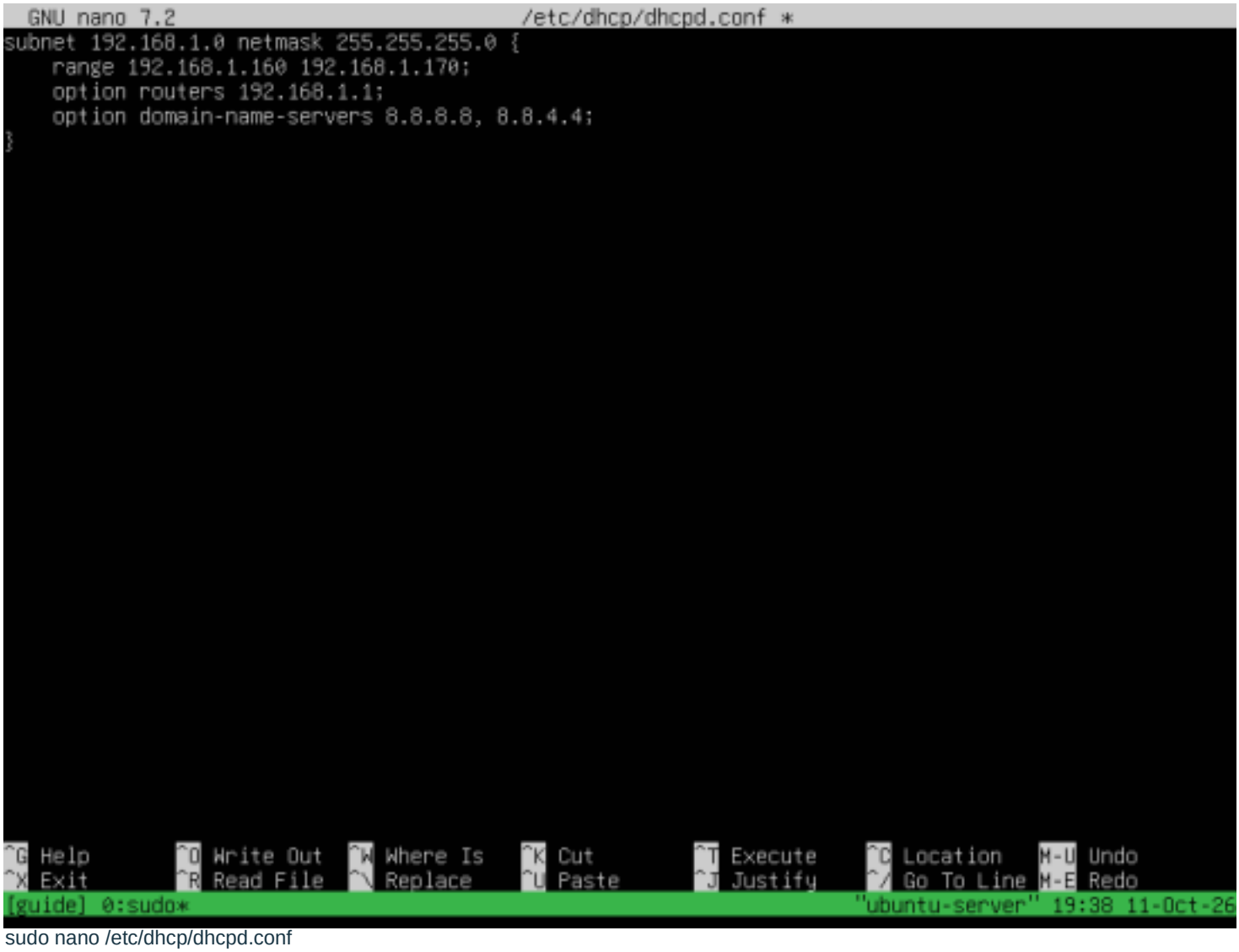
Part 2.1 - Install DHCP



sudo apt install isc-dhcp-server -y

The specified DHCP package installation command completed. Because isc-dhcp-server was already installed, apt kept the existing latest version instead of reinstalling it.

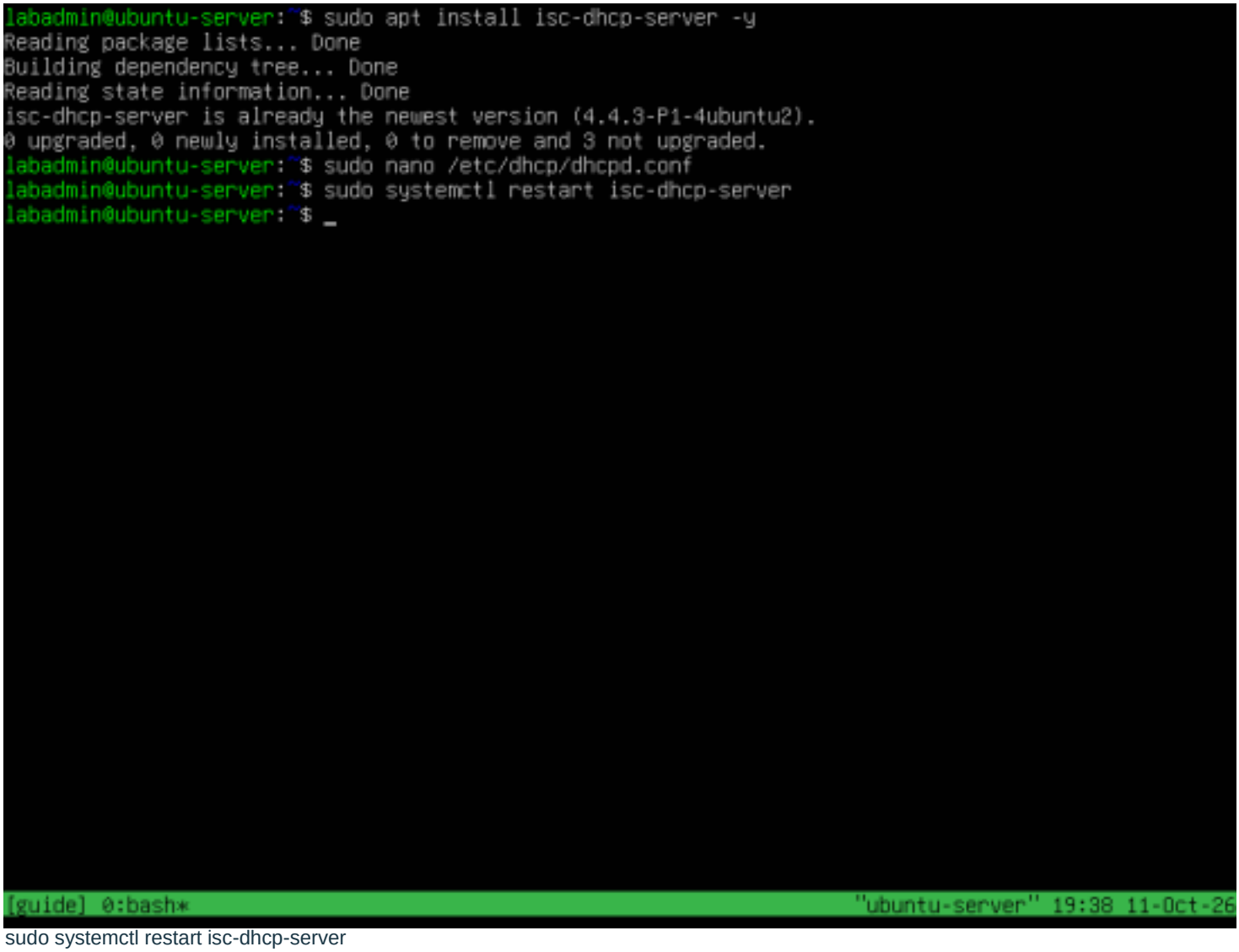
Part 2.2 - Configure DHCP



sudo nano /etc/dhcp/dhcpd.conf

The configuration uses subnet 192.168.1.0 with subnet mask 255.255.255.0. The address pool runs from 192.168.1.160 to 192.168.1.170, the router is set to 192.168.1.1, and the DNS servers are 8.8.8.8 and 8.8.4.4, as listed in the guide. The edited file was saved.

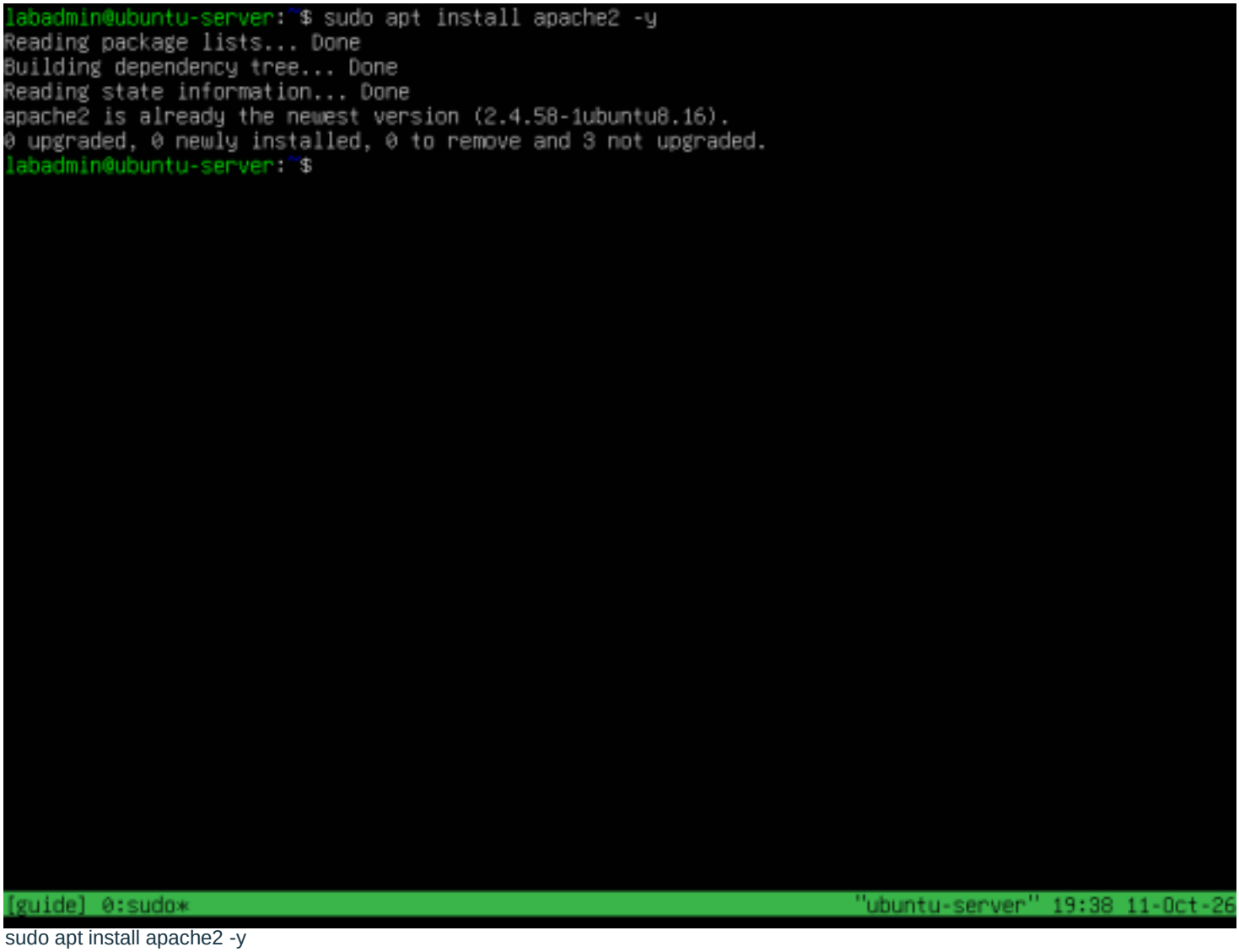
Part 2.3 - Restart DHCP



sudo systemctl restart isc-dhcp-server

The restart command returned to the terminal without an error. A separate service-status check showed that DHCP was active. This screenshot records the restart and status check only; this section of the guide did not require a client lease test.

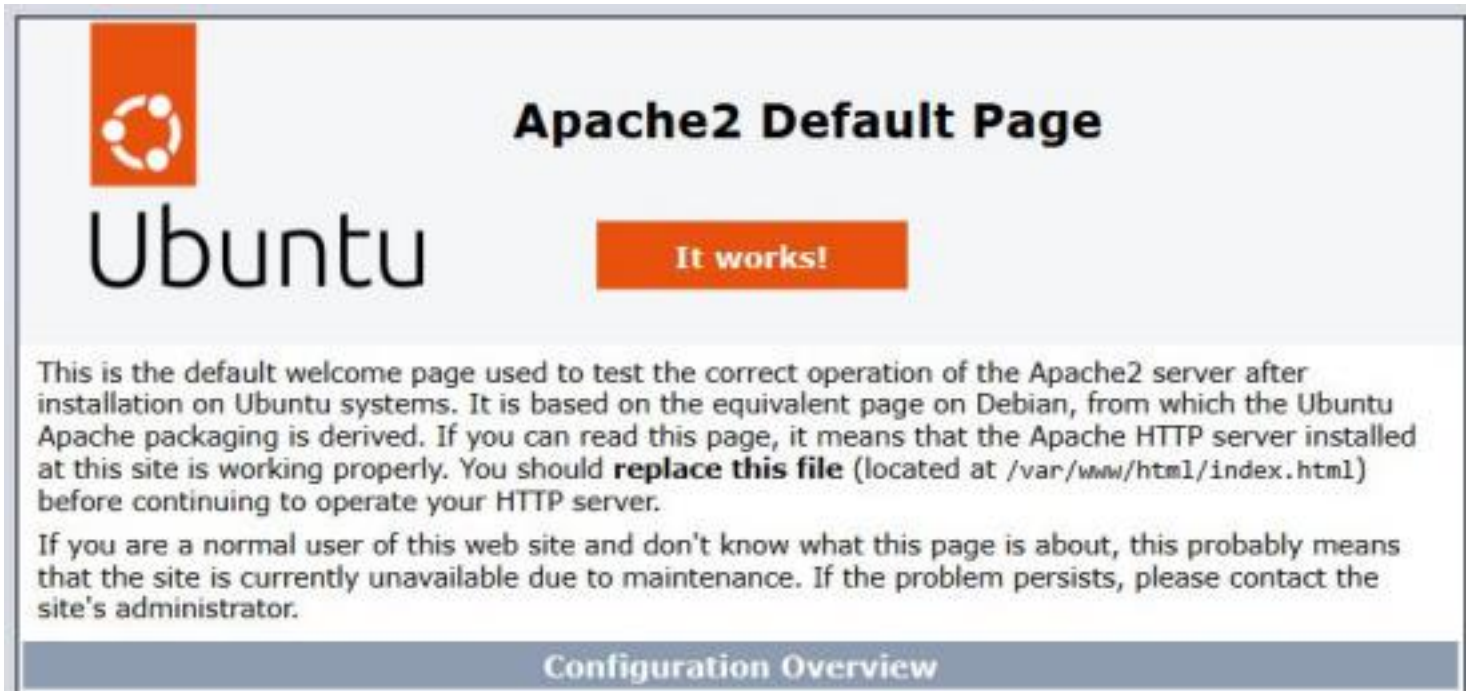
Part 3.1 - Install Apache



sudo apt install apache2 -y

The Apache installation command completed successfully. Since Apache was already installed, apt reported the version currently available on the system.

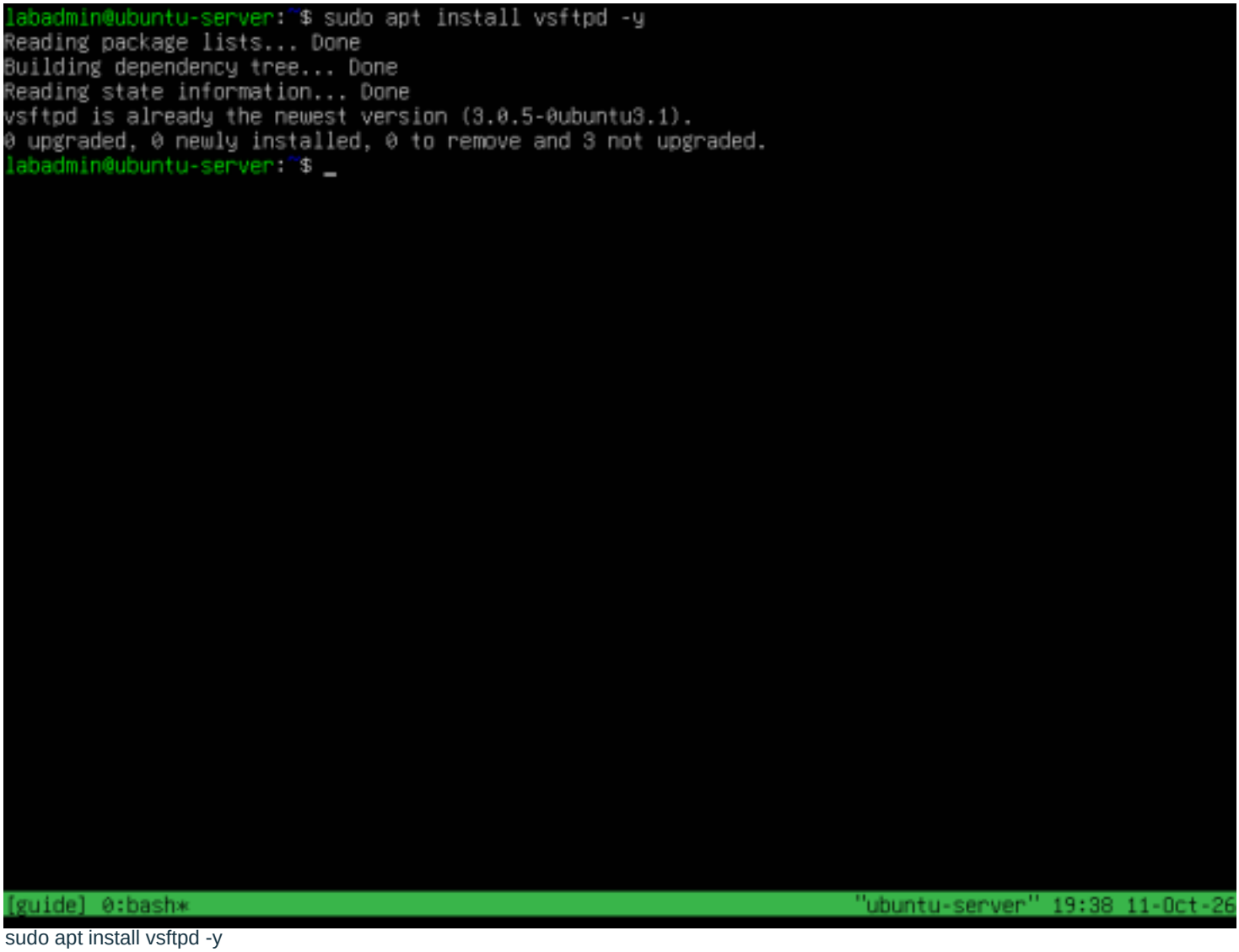
Part 3.2 - Open the host browser



http://192.168.1.150/

The Windows host browser was directed to the guide's address, http://192.168.1.150/, and displayed the Apache2 Default Page with the “It works!” message. The screenshot captures the top of the page, while the full browser capture is provided separately as 10-apache-browser.jpg.

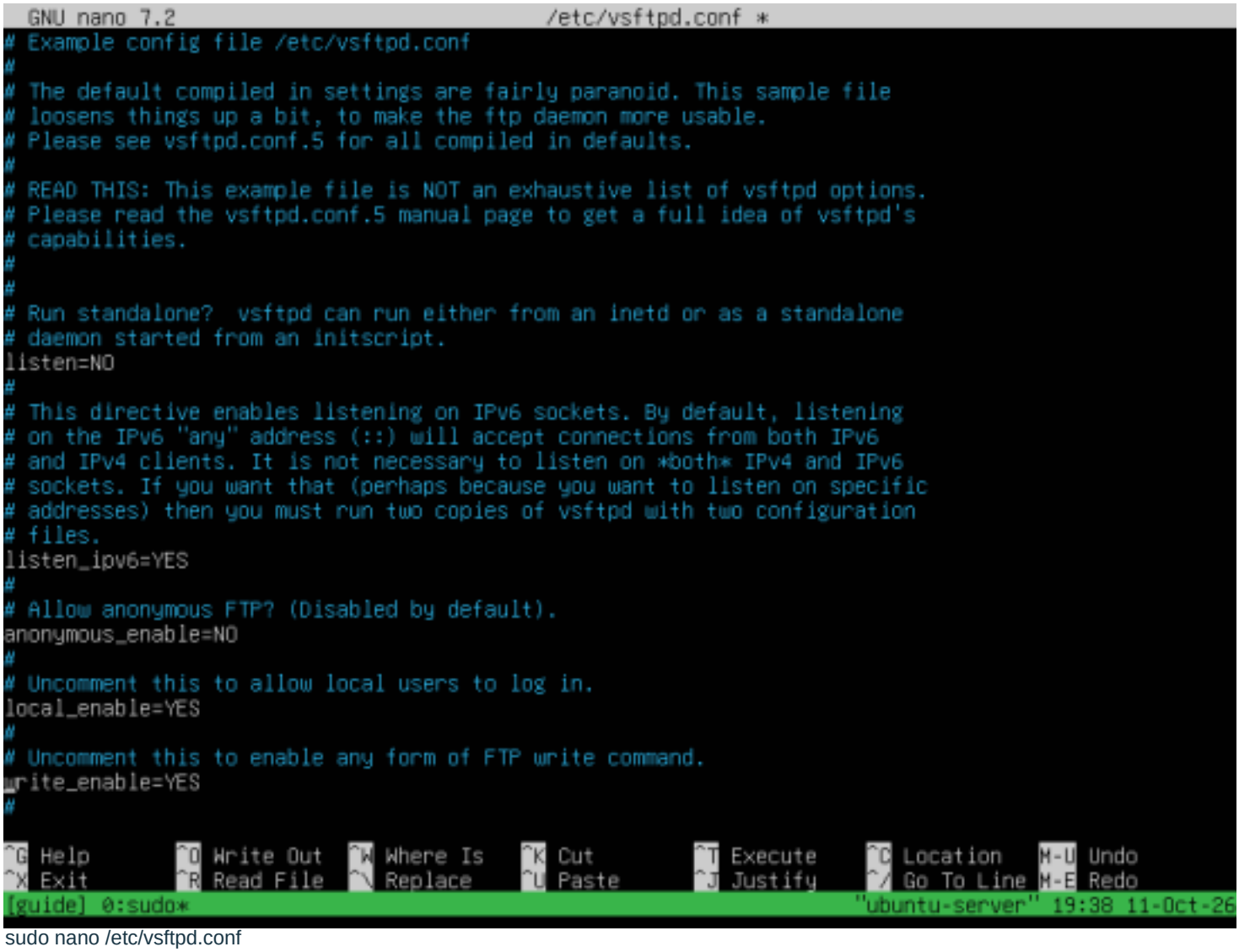
Part 4.1 - Install FTP



sudo apt install vsftpd -y

The requested vsftpd installation command completed. The package was already at the newest available version, so apt did not reinstall it.

Part 4.2 - Enable FTP writes



sudo nano /etc/vsftpd.conf

In nano, the existing write_enable=YES setting was uncommented and the configuration was saved. The screenshot shows the setting without a leading #. This allows FTP write commands when the logged-in user also has the required filesystem permissions.

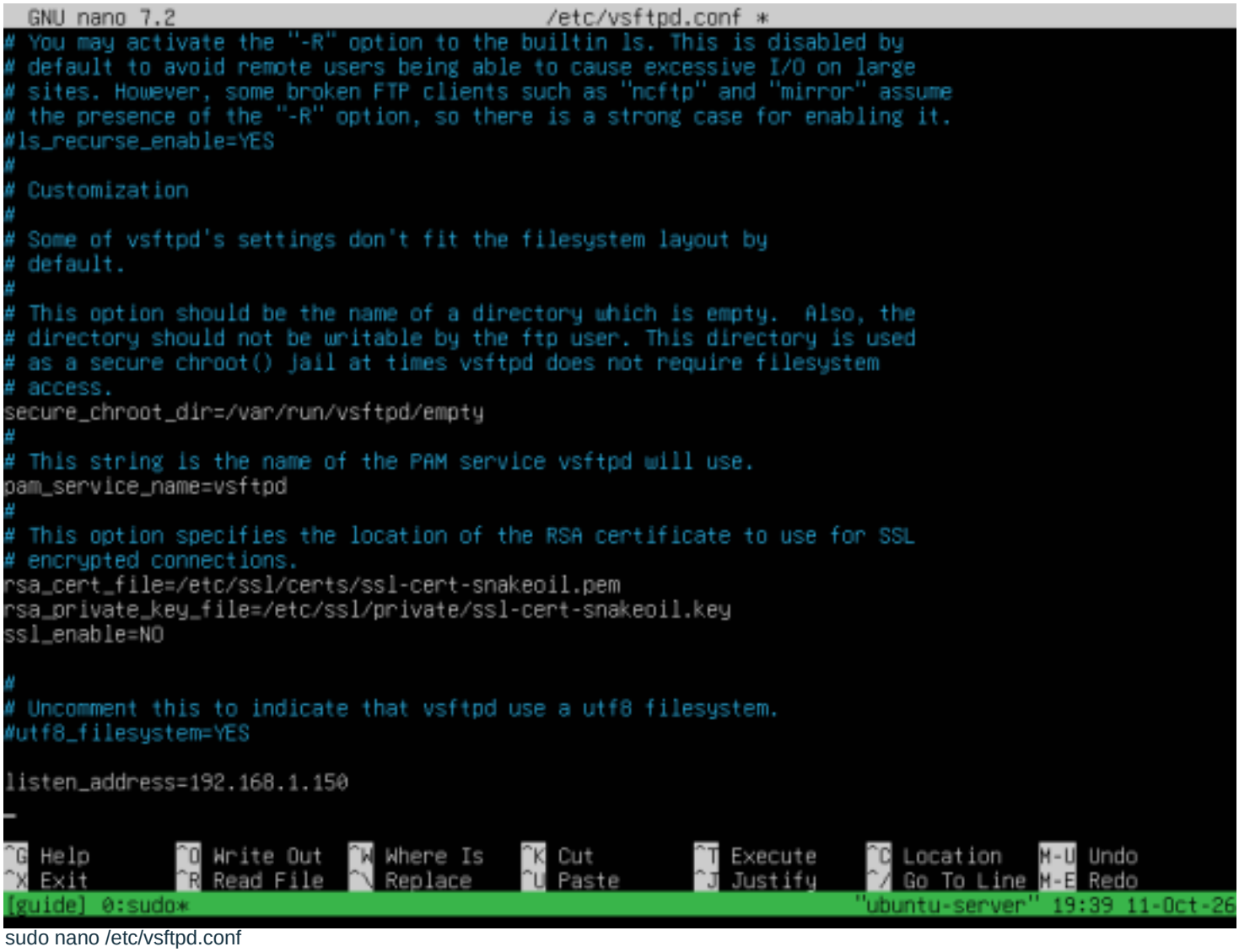
Part 4.3-4 - Restart and test FTP



sudo systemctl restart vsftpd
ftp 192.168.1.150

After vsftpd restarted, the command connected to 192.168.1.150. The existing ftpstudent account signed in successfully, and the server's 230 response confirmed authentication. The password was not shown on screen, and the session was ended with bye after the screenshot.

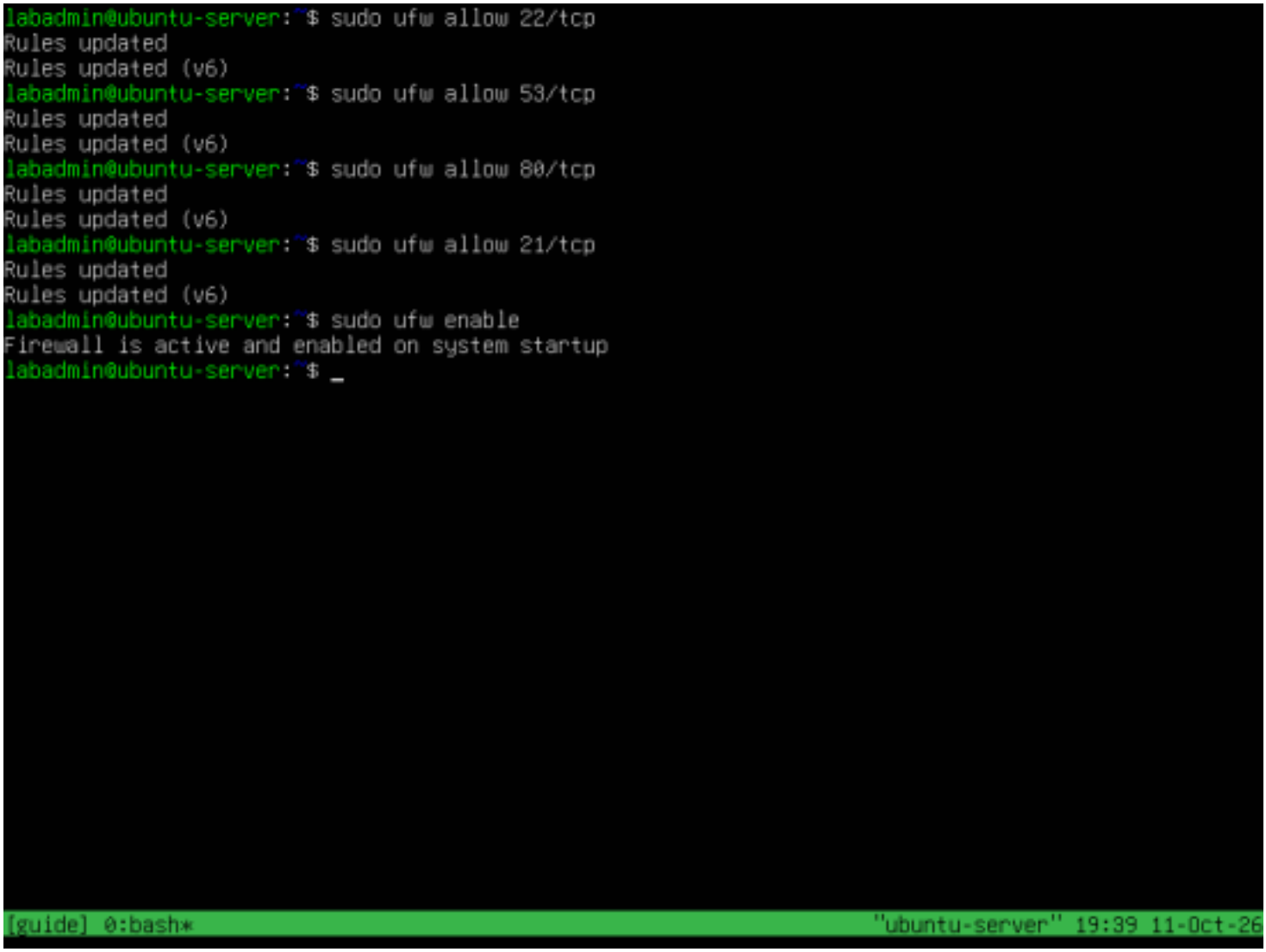
Part 5.1 - Add the listen address



sudo nano /etc/vsftpd.conf

The setting listen_address=192.168.1.150 was added to the configuration and saved. The screenshot confirms that the line is present, but it does not prove that access is restricted to a particular client network. The guide does not show a restart after this change.

Part 5.2 - Enable UFW



sudo ufw allow 22/tcp
sudo ufw allow 53/tcp
sudo ufw allow 80/tcp
sudo ufw allow 21/tcp
sudo ufw enable

The four allow commands were entered in the order shown, followed by ufw enable. The terminal output confirms that the firewall is active and configured to start automatically.

Part 5.3 - Verify the rules

sudo ufw status

The guide's final step asks for the rules to be checked but does not name a specific command. Running ufw status displayed the active firewall rules: TCP ports 22, 53, 80, and 21 are listed for both IPv4 and IPv6. No other allow rules were added.

What the results establish

The listed installation, configuration, restart, and firewall steps were completed. The local DNS lookup, Apache page check from the host browser, and FTP login also succeeded. BIND9, DHCP, Apache, and vsftpd were

active when their status was checked.

Firewall coverage: the guide opens TCP ports 22, 53, 80, and 21 only. It does not add UDP port 53 for standard remote DNS requests, UDP port 67 for DHCP, or a rule for FTP data transfers. Therefore, a successful local DNS lookup and FTP login do not confirm that these other network connections will work with UFW enabled.

FTP listening address: the file contains the added listen_address setting. However, the restored Ubuntu configuration has listen=NO and listen_ipv6=YES, and no restart is shown after the Part 5 edit. The line by itself does not restrict which client networks can connect. The UFW rules also apply to Anywhere rather than limiting traffic to the local subnet.

DHCP settings: the router value 192.168.1.1 and Google's DNS addresses were kept as specified in the guide. The host-only network is not connected to that router, so this report does not claim that clients can access the internet. The public DNS servers also do not host the private example.local zone.

Testing limits: this run did not include a DHCP lease demonstration or an FTP file-transfer test beyond the checks requested in the guide. Lease and transfer tests from the earlier report used a previous configuration, so they cannot be treated as proof for the current setup.

Included files: the archive contains this report, its screenshots, and an HTML step-by-step guide. The account passwords are kept separately in LOGIN-PRIVATE.txt, which is not part of the archive.